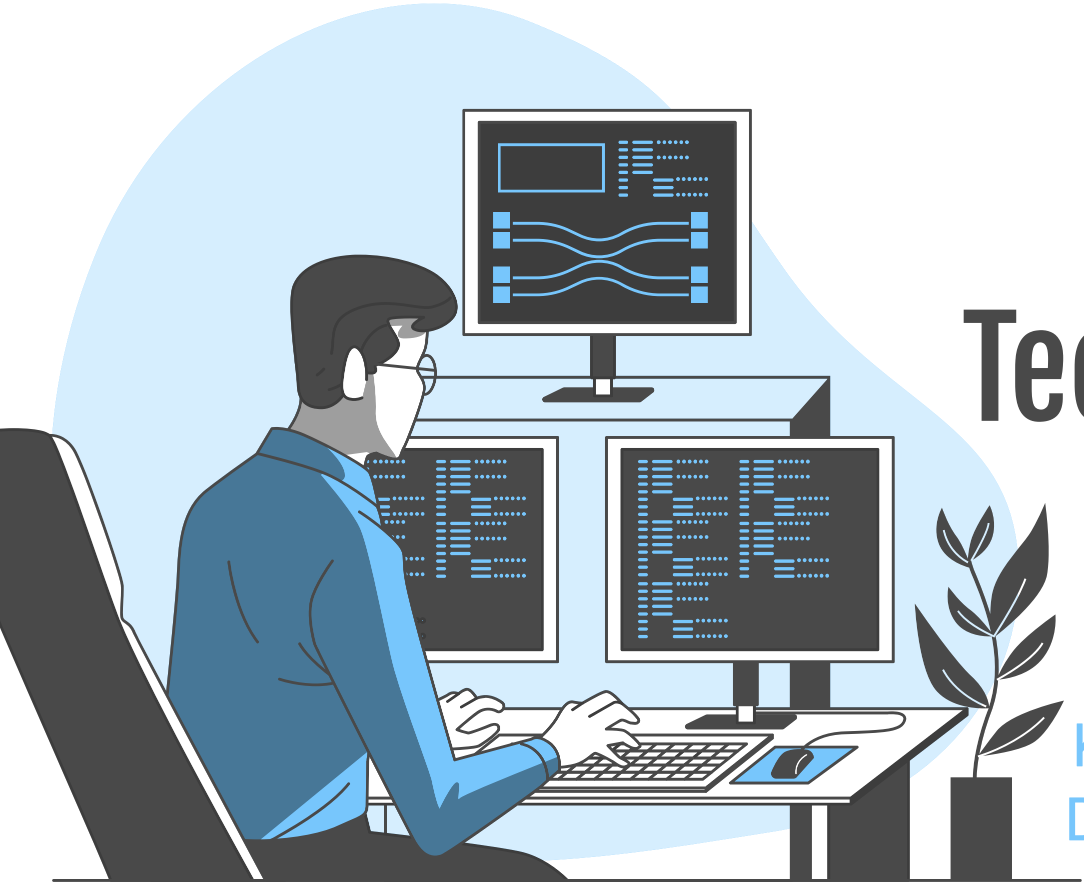

Tech Employment in the Age of AI

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Kaitlyn Chou, and Emily Friedman

Ds 4002 | Group 6 | March 27, 2026

Tech workers laid off since 2022

500k+

Tech workers laid off since 2022

Motivation and Context

Research Question

To what extent has the development and adoption of AI systems impacted employment levels in the global technology sector, and what do time-series modeling of 2020-2025 suggest about the likelihood of recovery in the next few years?

Hypothesis

The current rise of AI will be associated with an increase in layoffs in the 2026-27 years, and exceed layoff numbers seen in non-COVID, pre-AI years.

Why Does This Matter?

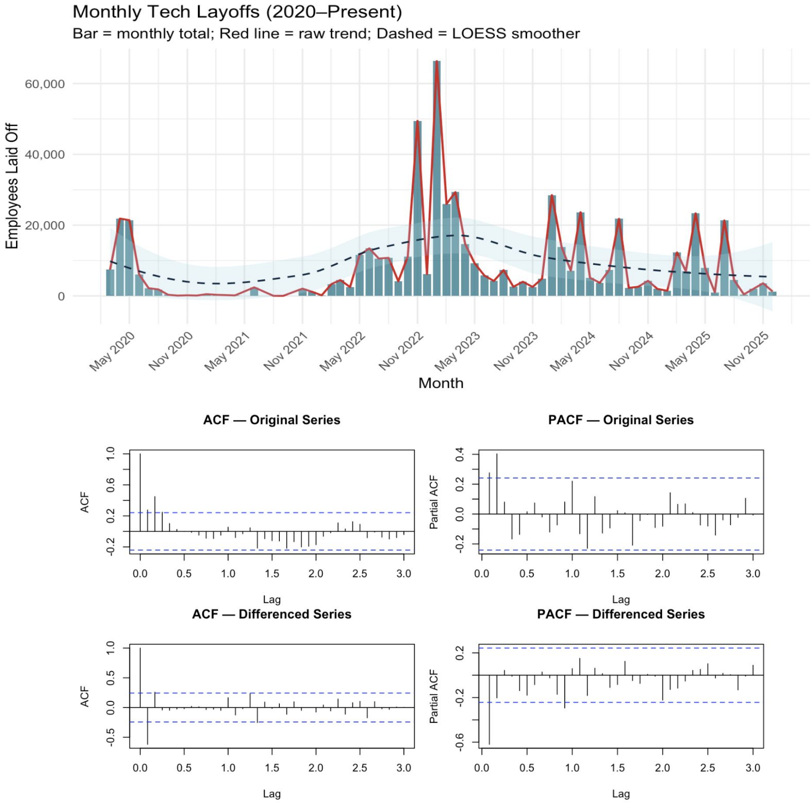
- ~7.4 million Americans are unemployed and actively job searching [2],
- For the first time since 2021, unemployed workers outnumber available jobs [3]
- Tracking layoffs helps reveal job market challenges, predict economic trends, and assess impacts on employee well-being [5]

Modeling Approach

SARIMA Model - Seasonal Autoregressive Integrated Moving Average

- Forecast time series data with trends and repeating seasonal patterns

Compare RMSE with naive seasonal baseline to see overall performance



Data Acquisition and Explanation

Dataset Source

- Layoff data collected from **Layoffs.fyi**, a publicly maintained tracker of global technology sector layoffs
- Aggregates publicly reported layoff announcements from company disclosures and news reports

Dataset Structure

- **Time range:** 2020 – 2025
- **Format:** Each row represents a single layoff event
- **Size:** Hundreds of rows covering major technology firms and startups

Data Acquisition and Explanation

Key Variables

- date – date of layoff announcement
- company – company conducting layoffs
- industry – sector within the tech ecosystem
- laid_off – number of employees affected

Processing for Analysis

- Data aggregated into monthly totals of employees laid off
- Location variables excluded from analysis to focus on global industry trends

Company	Location_HQ	Region	USState	Country	Continent	Laid_Off	Date_layoffs	Percentage	Company_Size_befc	Company_Size_afte	Industry	Stage	Money_Raised_in__	Year
Tamara Mellon	Los Angeles	other	California	USA	North America	20	2020-03-12	40	50	30	Retail	Series C	90	2020
HopSkipDrive	Los Angeles	other	California	USA	North America	8	2020-03-13	10	80	72	Transportation	Unknown	45	2020
Panda Squad	San Francisco	San Francisco Bay A	California	USA	North America	6	2020-03-13	75	8	2	Consumer	Seed	1	2020
Help.com	Austin	other	Texas	USA	North America	16	2020-03-16	100	16	0	Support	Seed	6	2020
Inspirato	Denver	other	Colorado	USA	North America	130	2020-03-16	22	591	461	Travel	Series C	79	2020
Flytedesk	Boulder	other	Colorado	USA	North America	4	2020-03-18	20	20	16	Marketing	Seed	4	2020
Remote Year	Chicago	other	Illinois	USA	North America	50	2020-03-19	50	100	50	Travel	Series B	17	2020
CTO.ai	Vancouver	Cascadia	British Columbia	Canada	North America	30	2020-03-20	50	60	30	Infrastructure	Seed	7	2020
Flywheel Sports	New York City	other	New York	USA	North America	784	2020-03-20	98	800	16	Fitness	Acquired	120	2020
Compass	New York City	other	New York	USA	North America	375	2020-03-23	15	2500	2125	Real Estate	Series G	1600	2020
Convene	New York City	other	New York	USA	North America	150	2020-03-23	18	833	683	Real Estate	Series D	280	2020
GrayMeta	Los Angeles	other	California	USA	North America	20	2020-03-23	40	50	30	Data	Unknown	7	2020
Ladder Life	Palo Alto	San Francisco Bay A	California	USA	North America	13	2020-03-23	25	52	39	Finance	Series C	94	2020
Leafly	Seattle	Cascadia	Washington	USA	North America	91	2020-03-23	50	182	91	Retail	Acquired	2	2020

Analysis Plan

01
Data Collection
- Source: layoffs.fyi
- Monthly Laid_Off totals
- Time range: 2020–2025

02
Pre-processing
- Drop location columns
- Remove missing rows
- Dickey-Fuller test
- Agg. to monthly totals

03
EDA & SARIMA
- STL: Q1 seasonality
- ACF/PACF: AR(1-2)
- d=1 differencing
- Structural break: 2023

04
Model Selection
- SARIMA(1,1,1)(1,1,0)
- auto.arima() AIC/BIC
- xreg: 2023 break
- Cross-validation applied

05
Forecast 2026–2027
- 24-month forecast
- Global monthly layoffs
- Train: 2020–2024
- Test: Jan–Dec 2025

06
Evaluation
- RMSE vs. naïve baseline
- Visualize forecast
- Goal: beat baseline
- Report findings

Goal: RMSE < seasonal naïve baseline · Forecast global monthly tech layoffs through 2027

Tricky Analysis Decision

Should we constrain auto.arima() to include seasonal components, even though it selects a non-seasonal model by default?

auto.arima (ARIMA)

- No seasonality captured — Q1 spikes ignored
- Higher AICc than constrained model
- Worse RMSE: 8,108 vs. naïve baseline 8,014
- MAE and MAPE substantially higher

SARIMA (Constrained)

- Captures Q1 seasonal spikes explicitly
- Lower AICc than auto.arima baseline
- Best RMSE: 7,745 — beats all competitors
- Ljung-Box $p = 0.94$: residuals ≈ white noise

Forcing seasonality reduced RMSE by ~4.5% vs. the naïve baseline. Without this constraint the model would have missed the recurring Q1 layoff spikes entirely.

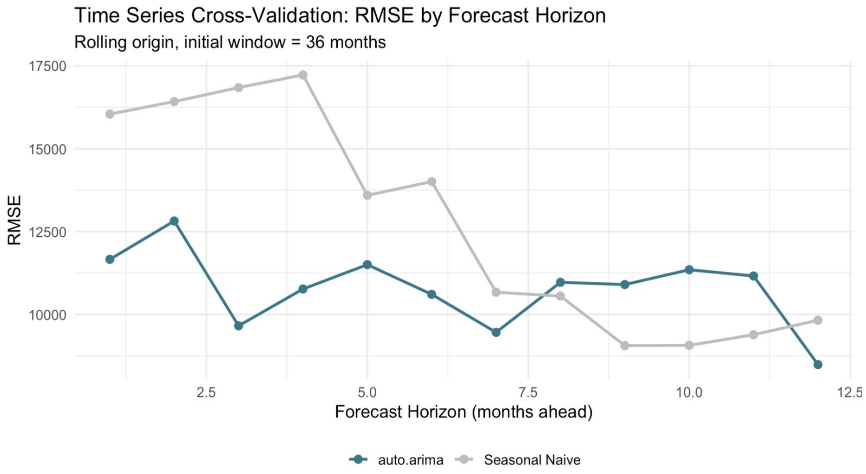
Bias and Uncertainty Validation

Model Validation:

- Residuals ≈ white noise (Ljung-Box $p = 0.94$)
- Model beats baseline (RMSE ↓ vs naive)

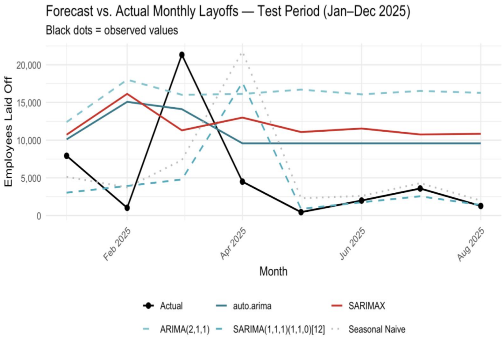
Limitations:

- Public reporting bias → underrepresents smaller firms and global markets
- 2023 spike is a one-time structural shock, not a repeatable pattern
- Short test period (8 months) limits evaluation of long-term accuracy



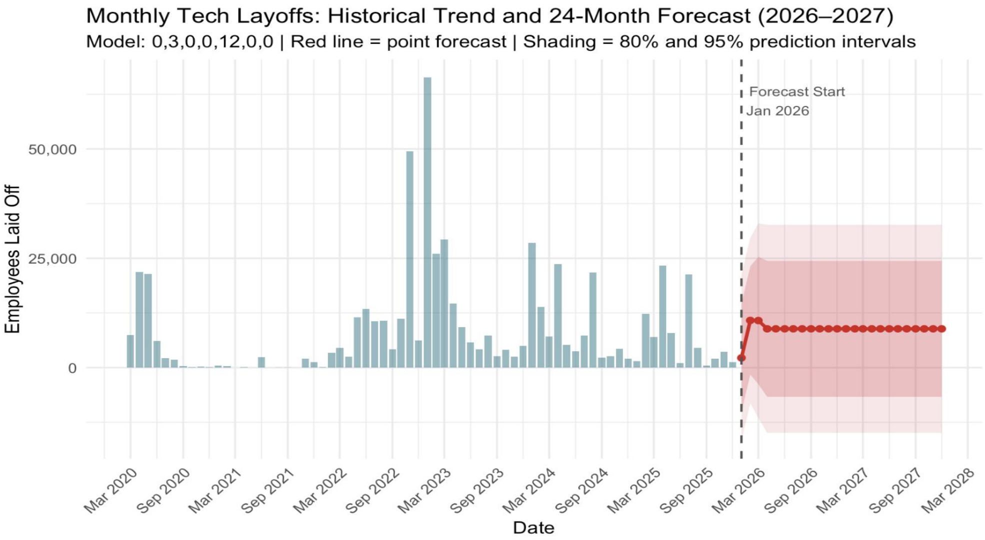
Results & Conclusions

- **Goal Achieved:** SARIMA beat the seasonal naive baseline
- **Forecast:** Layoffs stabilize around ~8.9K/month through 2027, with Q1 spikes
- AI restructuring has raised layoff baseline higher than pre-2022 levels, but the acute phase appears to be over
- **Hypothesis Partially Supported:** Layoffs won't hit 2023 peaks again, but won't return to 2021 (COVID) lows



Out-of-Sample Forecast Accuracy on Test Set (Jan–Dec 2025)

Model	RMSE	MAE	MAPE
SARIMA(1,1,1)(1,1,0)[12]	7745.41	4912.28	107.44
Seasonal Naive (Baseline)	8013.90	5076.88	158.03
auto.arima (no xreg)	8108.46	7442.32	596.11
SARIMAX (auto + break_2023)	9723.23	9169.66	691.55
ARIMA(2,1,1)	12883.02	12087.93	980.92



Next Steps

New Lines of Exploration

Look at additional external regressors (inflation, fiscal cycles)

Improvements

Utilize the full 2025 data (when it's available) for better RMSE estimates

Split data by industry and use separate models for each industry

New Questions

How well would this model generalize to other industries?

Are there any other factors that could contribute to the current pattern of layoffs?

References and Acknowledgements

Github Page: https://github.com/jensen-harvey/DS4002-project2.git

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[6] Layoffs. fyi, “Tech Layoffs.” [Online]. Available: https://layoffs.fyi/ [Accessed: Mar. 15, 2026.]

Questions?